

Predicting Used-Car Prices in UK Market with Supervised Machine Learning

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DATA 1030: Hands-on Data Science

GitHub Repository:

https://github.com/MurphyChen0413/data1030final_project

December 14, 2025

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1 Introduction

Used cars represent one of the largest segments of the automotive market in the UK and globally, yet the market remains highly opaque, with substantial price variation for similar vehicles across sellers and platforms. Prices depend on many interacting factors—such as mileage, vehicle age, brand reputation, and mechanical specifications—creating strong non-linear relationships that are difficult to capture with traditional statistical models. Accurate price prediction can improve transparency for consumers and support dealerships and on-line platforms in appraisal and inventory management. This project develops a supervised learning pipeline to predict used-car prices and identify the features that most strongly drive price variation.

The analysis uses the **DVM-CAR** dataset, a large public collection of UK used-car listings. After preprocessing, the dataset contains approximately 536,000 observations and 11 features, spanning over 900 models across 100 brands and vehicles registered between 2000 and 2020. Prior work shows that machine-learning methods—particularly tree-based ensembles—outperform traditional regression models in this setting, with reported R^2 values between 0.75 and 0.90 [Jin et al., 2023, Lin, 2025, Bukvić, 2022]. Building on this literature, the present study evaluates both linear and nonlinear models, applies systematic hyperparameter tuning, and uses interpretability techniques to explain how vehicle attributes influence price formation in the used-car market.

2 EDA

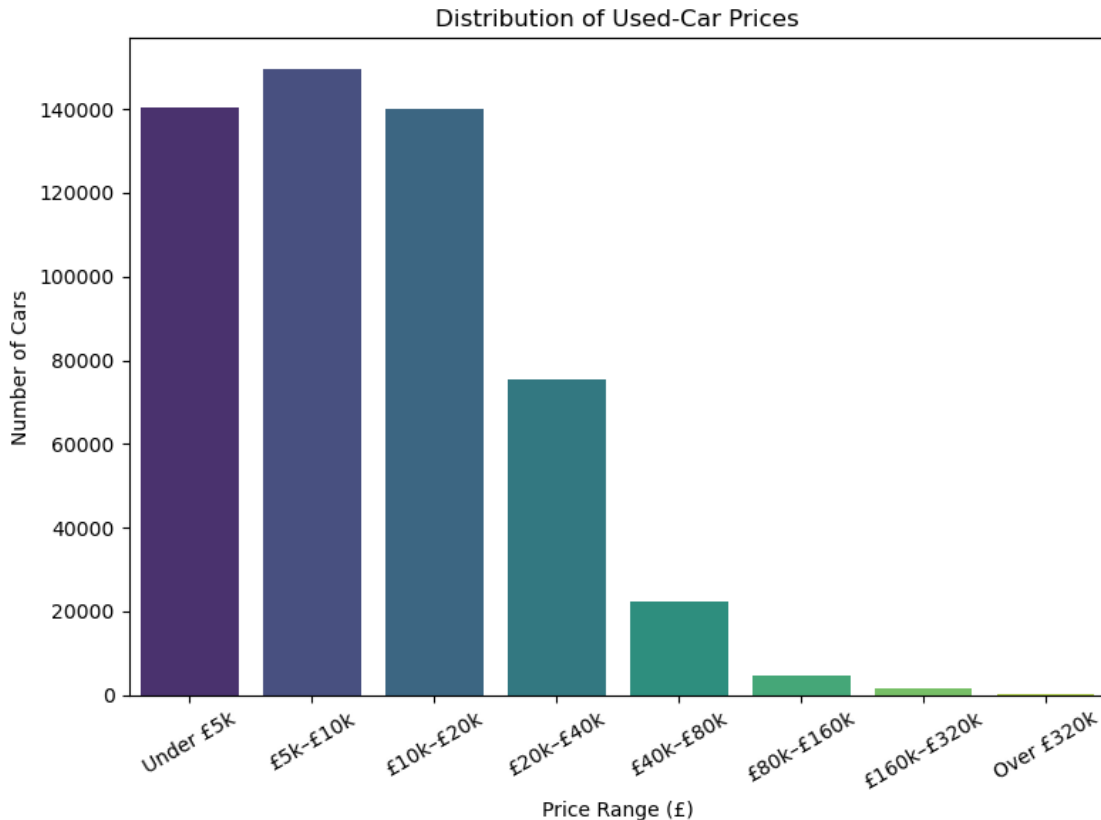


Figure 1: Distribution of used-car prices in the DVM-CAR dataset.

The distribution of used-car prices is highly uneven, with the majority of listings concentrated in the lower price ranges (Figure 1). More than half of all vehicles are priced below £20,000, particularly within the “Under £5k,” “£5k–£10k,” and “£10k–£20k” categories. The number of listings declines sharply for higher price groups, with luxury vehicles above £80,000 representing only a small fraction of the dataset. This long-tailed pattern reflects both the affordability structure of the used-car market and the wide variability in vehicle depreciation across brands and models.

The relationship between mileage and price shows a clear nonlinear depreciation pattern (Figure 2). Low-mileage vehicles cluster at higher prices, while prices decline rapidly with increasing mileage before flattening at higher mileage levels. Beyond roughly 80,000–100,000 miles, prices become compressed, with most vehicles converging toward lower price ranges regardless of additional mileage.

Body-type frequencies also reveal significant imbalance in the dataset (Figure 3). Hatchbacks dominate the UK used-car market, representing more than 200,000 listings—nearly double the number of SUVs, the second-largest category. The prevalence of certain body types reflects market preferences and contributes to systematic differences in pricing patterns captured later by the machine-learning models.

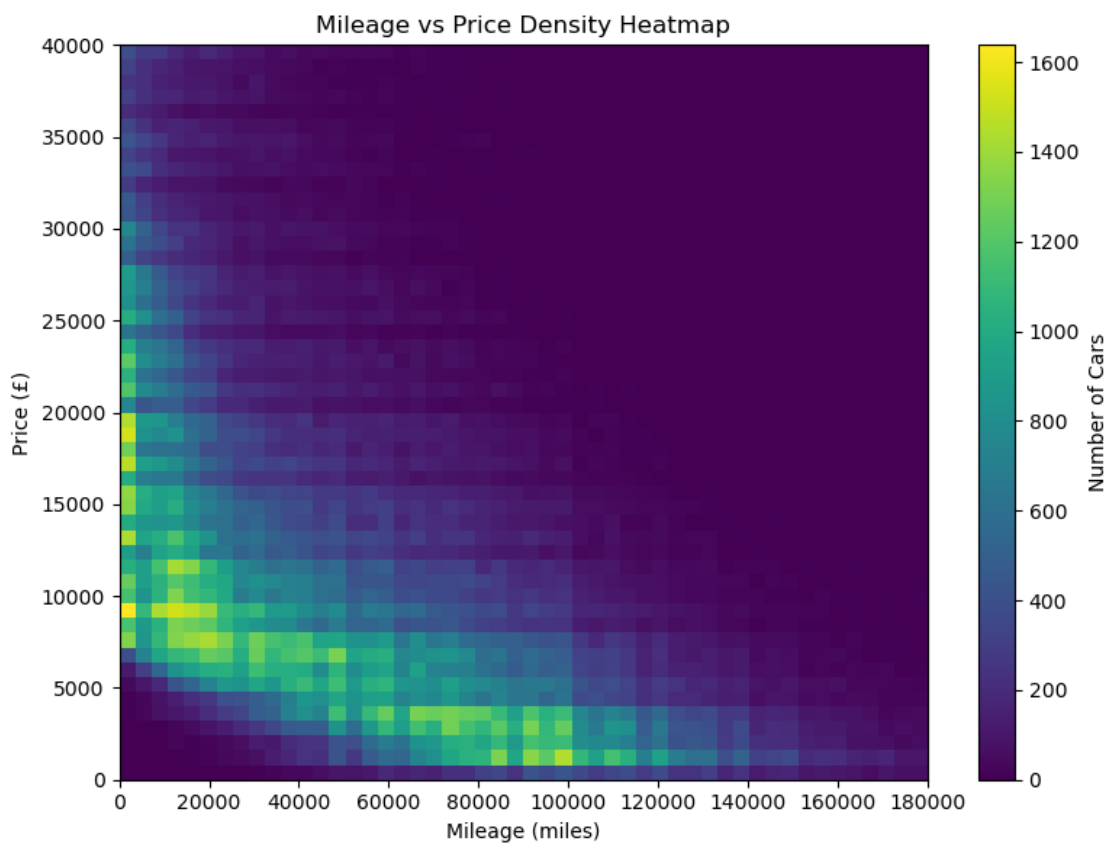


Figure 2: Density heatmap between mileage and price

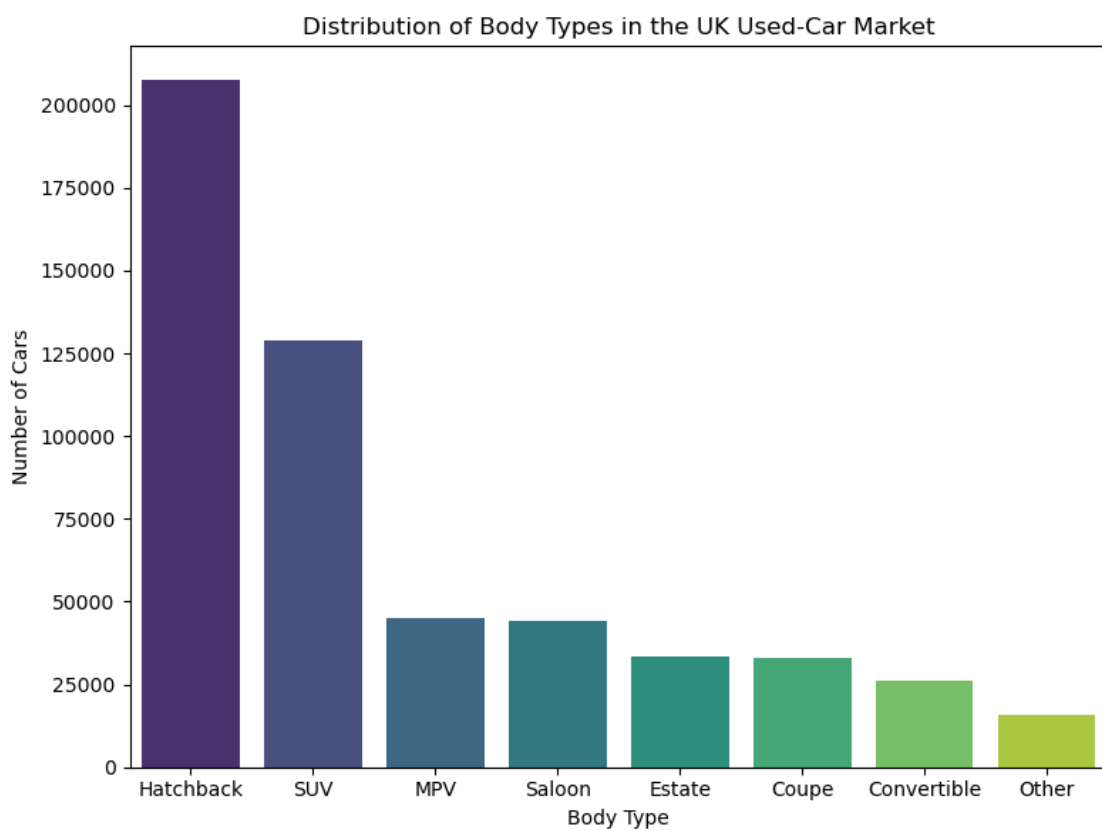


Figure 3: Distribution of body types in the DVM-CAR dataset.

3 Methods

3.1 Data Splitting Strategy

Because hyperparameter tuning was performed using `GridSearchCV`, which internally applies k -fold cross-validation, the dataset was split only once into a held-out test set and rest of the data remain in the Other group. Specifically, 20% of the data was reserved as the final test set, while the remaining 80% was used exclusively within the cross-validation framework for model training and hyperparameter selection. Using the full 80% of data for cross-validation increases the effective training size in each fold and leads to more reliable hyperparameter estimates. The held-out 20% test set was not used during any stage of model fitting or tuning and serves as an unbiased estimate of final model performance.

3.2 Preprocessing and Feature Engineering

Several preprocessing steps were applied to prepare the dataset for modeling. All transformations were implemented within scikit-learn pipelines to ensure that preprocessing was learned only from training folds during cross-validation, preventing information leakage. Basic cleaning standardized column formats, removed duplicate listings, and consolidated categorical labels to reduce fragmentation across variables such as body type, fuel type, gearbox, brand, and model. Missing categorical values were imputed as an explicit “Unknown” category and subsequently one-hot encoded. Numerical features were imputed using an `IterativeImputer` with a Random Forest estimator for linear models, while tree-based models such as XGBoost handled missing values natively through their split logic.

3.3 Machine Learning Pipeline

All models were implemented using scikit-learn pipelines to ensure that preprocessing and model fitting were performed consistently and without data leakage. The pipeline combined imputation, one-hot encoding, and the final estimator.

For linear model, numerical features were imputed using an `IterativeImputer` and standardized with `StandardScaler`, while categorical features were one-hot encoded. Tree-based models such as XGBoost used the same categorical encoder but operated on unscaled numerical inputs, because it XGBoost has its built-in way of handling missing values.

Hyperparameter tuning and model selection were implemented inside `GridSearchCV`, which applied 3-fold cross-validation on the 80% training portion of the data. The best model from cross-validation was then refit on the full training data and evaluated once on the held-out 20% test set.

3.4 Evaluation Metrics Interpretability

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). All metrics were computed on the held-out 20% test set to provide an unbiased estimate of generalization performance.

Model interpretability was assessed entirely using SHAP, which provides both global and local insight into how the model uses each feature to generate predictions. For global interpretability, SHAP summary plots were computed from the XGBoost model, which achieved the strongest predictive performance. For local interpretability, SHAP waterfall plots were used to explain individual predictions, illustrating precisely how each feature pushes a car’s predicted price upward or downward relative to the model’s expected value.

3.5 Hyperparameter Tuning

For the Elastic Net regression model, two hyperparameters were tuned: the overall regularization strength and the mixing ratio between L1 and L2 penalties. The regularization parameter was varied on a logarithmic grid from 10^{-3} to 10^2 , and the L1 ratio was selected from $\{0.1, 0.5, 0.9\}$.

For the Random Forest, the grid focused on the number and depth of trees, along with leaf size: the number of trees was set to $\{200, 400\}$, the maximum depth to $\{None, 20, 40\}$, and the minimum number of samples per leaf to $\{1, 5\}$. These parameters control the bias–variance trade-off and were chosen analogously to the random forest grids used in class.

For XGBoost, the hyperparameter grid focused on the key parameters that control model complexity and regularization. The grid varied the learning rate $\{0.01, 0.1, 0.2\}$, maximum tree depth $\{3, 5, 7\}$, and the minimum child weight $\{1, 3, 5\}$, which jointly determine the size and smoothness of individual boosting trees. Sampling-based regularization was explored through subsampling rates $\{0.7, 0.8, 0.9\}$ and column-subsampling rates $\{0.7, 0.8, 0.9\}$, while the number of boosting rounds was tuned over $\{100, 500, 1000\}$.

For the KNN, the number of neighbors was tuned over $\{5, 10, 20\}$, the weighting scheme over $\{uniform, distance\}$, the Minkowski distance parameter p over $\{1, 2\}$, and the leaf size over $\{20, 40\}$. These parameters jointly determine how local the model is and how distances between observations are computed.

3.6 Uncertainty Estimation

To assess the robustness of the evaluation metrics, two sources of uncertainty were examined. First, uncertainty due to data splitting was measured by repeating the train-test split across five different random seeds and re-evaluating a baseline model. The variation in MAE across these splits provided an estimate of how sensitive performance is to the particular partition of the data.

Second, uncertainty from nondeterministic learning algorithms was evaluated by refitting the Random Forest and XGBoost models multiple times on the same split. Because these models rely on bootstrap sampling and subsampling, their predictions vary slightly with each run. The resulting spread in MAE values was small, indicating that both models are stable under repeated training.

4 Results

4.1 Predictive Performance

Table 1 summarizes the test-set performance of all models. The baseline model, which always predicts the mean training price, performs poorly with an MAE of £10,972 and an RMSE of £23,209, corresponding to $R^2 = 0$. This provides a reference for the difficulty of the task and the amount of variability in used-car prices.

Table 1: Test-set performance of all models. Lower MAE and RMSE indicate better performance.

Model	MAE (£)	RMSE (£)	R^2
Baseline	10,972.00	23,209.00	0.0000
XGBoost	1,239.82	2,714.37	0.9863
Elastic Net	5,896.75	15,912.27	0.5299
Random Forest	1,029.52	5,325.80	0.9473
KNN	1,762.00	2,496.02	0.9884

All learned models substantially outperform the baseline. Elastic Net reduces the MAE to about £5,897 and achieves $R^2 \approx 0.53$, indicating that a regularized linear model can capture a sizable portion of the variation in prices but struggles with the strong nonlinearities observed in the EDA. The nonlinear models perform markedly better. Random Forest reaches an MAE of roughly £1,030 and $R^2 \approx 0.95$, reflecting its ability to model nonlinear effects and interactions among mileage, car age, engine size, and categorical attributes.

XGBoost and KNN obtain the strongest overall scores. XGBoost achieves an MAE of £1,240 and an RMSE of £2,714 with $R^2 = 0.9863$, while KNN attains a slightly lower RMSE and marginally higher R^2 but a larger MAE (around £1,762). Based on five repeated train-test splits, the MAE of the baseline model varies only modestly, so the improvement of XGBoost over the baseline corresponds to many standard deviations above the baseline MAE. In practice, XGBoost is preferred as the final model because it offers a better balance between error magnitude, stability, and interpretability.

4.2 Global Feature Importance

Global feature importance was evaluated using three methods: XGBoost’s built-in feature importance, permutation importance, and SHAP. First, XGBoost feature importance scores based on gain were examined to identify which features most frequently contribute to reducing loss during tree splits. Second, permutation importance was computed on the test set by measuring the increase in MAE when individual features were randomly shuffled. This approach provides a model-agnostic assessment of how much each feature contributes to predictive accuracy. Third, global SHAP summary plots were used to aggregate absolute SHAP values across observations.

Across all three methods, engine size, car age, and mileage consistently emerge as the most influential predictors of used-car prices. Larger engines are associated with positive contributions to predicted price, while higher mileage and older vehicle age exert strong

negative effects. Among categorical variables, gearbox type and body type are highly influential: automatic gearboxes and SUVs tend to increase predicted prices. Features such as door number and several infrequent body types exhibit consistently low importance across all methods, indicating that they contribute little to price prediction once key attributes such as mileage, age, engine size, and brand are taken into account. Overall, the close agreement among XGBoost feature importance, permutation importance, and SHAP indicates that the model relies on intuitive and economically meaningful drivers of used-car prices.

4.3 Local Explanations with SHAP

Local feature importance was examined using SHAP waterfall plots for individual listings. For a representative high-priced SUV, the base price prediction is pushed upward by a combination of large engine size, relatively low mileage, recent registration year, SUV body type, and a premium brand. Negative contributions arise from factors such as a manual gearbox or less popular color choices when present. In contrast, for a low-priced economy hatchback with high mileage, the waterfall plot shows large negative contributions from car age and mileage, with only modest positive contributions from engine size or favorable body type. These local explanations align well with domain intuition: the model prices cars higher when they are newer, less driven, larger-engined, and associated with premium brands, and lower when they are older, heavily used, and associated with budget segments.

5 Outlook

Several extensions could further improve predictive performance. From a modeling perspective, expanding the hyperparameter search space—such as tuning deeper trees, wider regularization ranges, and alternative learning rates—may yield additional gains. Evaluating a broader set of models covered could provide better comparison to choose a most optimal one. Further feature engineering, such as interaction terms between brand and body type or mileage and age, and the use of outlier-robust training strategies, may also improve model stability and accuracy.

Improvements in data quality would likely lead to better predictions. Incorporating data from additional markets and expanding the feature set to include variables such as vehicle condition, accident history, clean-title status, and seller characteristics could reduce unexplained variance and enhance the reliability of price estimates in real-world applications.

References

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